H5: Magnetostatics

5.1 the Lorentz force law

5.1.2 magnetic forces

Lorentz force law

The magnetic force on a charge Q with velocity v in a magnetic field is:

$$\mathbf{F}_{\text{mag}} = Q(\mathbf{v} \times \mathbf{B}).$$

or thus in an E and B field:

$$\mathbf{F} = Q[\mathbf{E} + (\mathbf{v} \times \mathbf{B})].$$

Work of magnetic force

The magnetic force cant do work

nl: if Q moves an amount dl = vdt, the work done is:

$$dW_{\text{mag}} = \mathbf{F}_{\text{mag}} \cdot d\mathbf{l} = Q(\mathbf{v} \times \mathbf{B}) \cdot \mathbf{v}\, dt = 0.$$

5.1.3 currents

current I

= charge per unit time passing a given point

> measured in Ampère: 1A = 1C/s

vb: for a line charge λ traveling through a wire at speed v we have: $\mathbf{I} = \lambda \mathbf{v}$.

force on a current

We have: $\mathbf{F}_{\text{mag}} = \int (\mathbf{v} \times \mathbf{B})\, dq = \int (\mathbf{v} \times \mathbf{B})\lambda\, dl = \int (\mathbf{I} \times \mathbf{B})\, dl.$

more general:

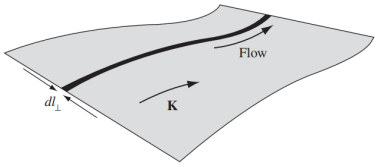
$$\mathbf{F}_{\text{mag}} = \int I\,(d\mathbf{l} \times \mathbf{B}).$$

surface current density K

= charge flow over a surface

> consider a ribbon of infinitesimal width $dl_\perp$ running parallel to the flow

> if the current in this ribbon is d**I** we have: $\mathbf{K} \equiv \frac{d\mathbf{I}}{dl_\perp}.$



force on K

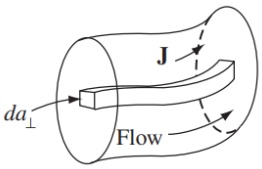
if the surface has charge density σ with velocity **v**, then **K**=σ**v**

> the magnetic force is: $\mathbf{F}_{\text{mag}} = \int (\mathbf{v} \times \mathbf{B})\sigma\, da = \int (\mathbf{K} \times \mathbf{B})\, da.$

volume current density J

for a tube of infinitesimal cross section $da_\perp$ running parallel to the flow

> the current in the tube is d**I** and we have: $\mathbf{J} \equiv \frac{d\mathbf{I}}{da_\perp}$



force on J

if the volue charge density is ρ with velocity **v**, then **J**=ρ**v** and we have:

$$\mathbf{F}_{\text{mag}} = \int (\mathbf{v} \times \mathbf{B})\rho\, d\tau = \int (\mathbf{J} \times \mathbf{B})\, d\tau.$$

continuity equation

The current crossing a surface S can be written as:

$$I = \int_S J\, da_\perp = \int_S \mathbf{J} \cdot d\mathbf{a}.$$

Then the charge per unit time leaving a volume V is:

$$\oint_S \mathbf{J} \cdot d\mathbf{a} = \int_V (\nabla \cdot \mathbf{J})\, d\tau.$$

since charge is conserved, whatever flows through the surface must come at an expense:

$$\int_V (\nabla \cdot \mathbf{J})\, d\tau = -\frac{d}{dt}\int_V \rho\, d\tau = -\int_V \left(\frac{\partial \rho}{\partial t}\right) d\tau.$$

thus we have:

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t}.$$

5.2 the Biot-Savart law

5.2.1 steady currents

magnetostatics

= steady currents produce magnetic fields that are constant in time:

Stationary charges ⇒ constant electric fields: electrostatics.
Steady currents ⇒ constant magnetic fields: magnetostatics.

thus electro/magnetostatics is the regime: $\frac{\partial \rho}{\partial t} = 0, \quad \frac{\partial \mathbf{J}}{\partial t} = \mathbf{0},$

5.2.2 the magnetic field of a steady current

Biot-Savart law

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{I} \times \hat{\boldsymbol{\imath}}}{\imath^2} dl' = \frac{\mu_0}{4\pi} I \int \frac{d\mathbf{l}' \times \hat{\boldsymbol{\imath}}}{\imath^2}.$$

> μ_0 = permeability of free space = $4\pi.10^{-7}$ N/A²
units: Tesla: 1T = 1N/(A.m)

Surface and volume Biot-Savart

for a surface: $\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{K}(\mathbf{r}') \times \hat{\boldsymbol{\imath}}}{\imath^2} da'$

for a volume: $\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}') \times \hat{\boldsymbol{\imath}}}{\imath^2} d\tau',$

5.3 the divergence and curl of B

5.3.1 straight-line currents

integral for of B

Consider an infinite straight wire
> the integral of B around a closed circular path with radius s is:

$$\oint \mathbf{B} \cdot d\mathbf{l} = \oint \frac{\mu_0 I}{2\pi s} dl = \frac{\mu_0 I}{2\pi s} \oint dl = \mu_0 I.$$

this is independent from the radius s
> nl: the magnetic field **B** decreases the same amount as s increases

Now consider a bundle of these wires
> each wire contributes $\mu_0 I$:

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}},$$

curl of **B**

Uses Stokes' theorem and rewrite $I_{enc} = \int \mathbf{J}.d\mathbf{a}$ to find:

$$\int (\nabla \times \mathbf{B}) \cdot d\mathbf{a} = \mu_0 \int \mathbf{J} \cdot d\mathbf{a},$$

hence:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}.$$

for infinite straight wires

5.3.2 the divergence and curl of B

divergence of **B**

Biot-Savart: $\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}') \times \hat{\boldsymbol{\imath}}}{\imath^2} d\tau'.$

apply the divergence: $\nabla \cdot \mathbf{B} = \frac{\mu_0}{4\pi} \int \nabla \cdot \left(\mathbf{J} \times \frac{\hat{\boldsymbol{\imath}}}{\imath^2} \right) d\tau'.$

where we now use: $\nabla \cdot \left(\mathbf{J} \times \frac{\hat{\boldsymbol{\imath}}}{\imath^2} \right) = \frac{\hat{\boldsymbol{\imath}}}{\imath^2} \cdot (\nabla \times \mathbf{J}) - \mathbf{J} \cdot \left(\nabla \times \frac{\hat{\boldsymbol{\imath}}}{\imath^2} \right)$

with $\nabla \times \mathbf{J} = 0$ since **J** doesn't depend on the unprimed variables
$\nabla \times (\boldsymbol{\imath}/\imath^2) = 0$

thus we have: $\nabla \cdot \mathbf{B} = 0.$

divergence of **B**	apply the curl to Biot-Savart: $$\nabla \times \mathbf{B} = \frac{\mu_0}{4\pi} \int \nabla \times \left(\mathbf{J} \times \frac{\hat{\boldsymbol{\imath}}}{\imath^2} \right) d\tau'.$$ where we can use: $\nabla \times \left(\mathbf{J} \times \frac{\hat{\boldsymbol{\imath}}}{\imath^2} \right) = \mathbf{J} \left(\nabla \cdot \frac{\hat{\boldsymbol{\imath}}}{\imath^2} \right) - (\mathbf{J} \cdot \nabla) \frac{\hat{\boldsymbol{\imath}}}{\imath^2}.$ for which it holds that: $\nabla \cdot \left(\frac{\hat{\boldsymbol{\imath}}}{\imath^2} \right) = 4\pi \delta^3(\boldsymbol{\imath}).$ and the second term is zero ** thus we find: $$\nabla \times \mathbf{B} = \frac{\mu_0}{4\pi} \int \mathbf{J}(\mathbf{r}') 4\pi \delta^3(\mathbf{r} - \mathbf{r}')\, d\tau' = \mu_0 \mathbf{J}(\mathbf{r}),$$ We have what we already found: $$\nabla \times \mathbf{B} = \mu_0 \mathbf{J},$$ ** because the derivative acts on $\boldsymbol{\imath}/\imath^2$ we can switch ∇ to -∇' $$-(\mathbf{J} \cdot \nabla) \frac{\hat{\boldsymbol{\imath}}}{\imath^2} = (\mathbf{J} \cdot \nabla') \frac{\hat{\boldsymbol{\imath}}}{\imath^2}.$$ now in particular the x-component is: $$(\mathbf{J} \cdot \nabla') \left(\frac{x - x'}{\imath^3} \right) = \nabla' \cdot \left[\frac{(x - x')}{\imath^3} \mathbf{J} \right] - \left(\frac{x - x'}{\imath^3} \right) (\nabla' \cdot \mathbf{J})$$ for steady currents **J**=0 thus: $$\left[-(\mathbf{J} \cdot \nabla) \frac{\boldsymbol{\imath}}{\imath^2} \right]_x = \nabla' \cdot \left[\frac{(x - x')}{\imath^3} \mathbf{J} \right]$$ and therefore this contribution to the integral can be written as $$\int_{\mathcal{V}} \nabla' \cdot \left[\frac{(x - x')}{\imath^3} \mathbf{J} \right] d\tau' = \oint_S \frac{(x - x')}{\imath^3} \mathbf{J} \cdot d\mathbf{a}'.$$ but we are integrating over the volume that appears in the Biot-Savart law > this is large enough to fit all the current, but we can make it arbitrarily big > this term will add nothing to the integral ie: on the boundary the current is zero > the surface integral vanishes, as we have used in out proof
5.3.3 Ampère's law	
Ampères law	$\nabla \times \mathbf{B} = \mu_0 \mathbf{J},$
integral Ampères law	We can change this formula via Stokes' theorem to: $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}.$
5.3.4 comparison of magnetostatics and electrostatics	
Maxwell's equations	We have four laws: two in electrostatics and two in magnetostatics: $\begin{cases} \nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0} \rho, & \text{(Gauss's law);} \\ \nabla \times \mathbf{E} = \mathbf{0}, & \text{(no name).} \end{cases}$ $\begin{cases} \nabla \cdot \mathbf{B} = 0, & \text{(no name);} \\ \nabla \times \mathbf{B} = \mu_0 \mathbf{J}, & \text{(Ampère's law).} \end{cases}$ we also have: F = Q(E + vxB)

5.4 the magnetic vector potential	
5.4.1 the vector potential	
vector potential A	Since ∇B=0, we can create a vector potential for which: $\mathbf{B} = \nabla \times \mathbf{A}.$
A in terms of J	for which we know that ∇(∇xA)=0 and Ampère's law: $\nabla \times \mathbf{B} = \nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} = \mu_0 \mathbf{J}.$ we can also find that: $\nabla \cdot \mathbf{A} = 0.$ which gives us: $\nabla^2 \mathbf{A} = -\mu_0 \mathbf{J}.$ or thus in integral form: $\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}')}{\imath} d\tau'.$

